Kongzi: A Historical Large Language Model with Fact Enhancement

Anonymous submission

Abstract

Large Language Models (LLMs) have achieved significant progress in natural language understanding and complex reasoning tasks. However, inaccuracies they often exhibit factual inaccuracies during long-chain reasoning or under misleading prompts, severely limiting their applicability in the historical domain, a field characterized by high information density and strict requirements for factual accuracy. In order to address these challenges, we propose **Kongzi**, a large language model specifically designed for historical understanding and complex reasoning. By integrating historical corpora and introducing a novel fact-enhanced reinforcement learning strategy, Kongzi significantly improves factual consistency and reasoning depth. Extensive evaluations of tasks such as historical question answering and counterfactual reasoning demonstrate that Kongzi consistently surpasses existing models even those with comparable or larger parameter scales in terms of factual accuracy and interpretative soundness.By effectively tackling the unique challenges inherent in historical texts, Kongzi sets a new benchmark for building accurate and reliable LLMs in knowledge-intensive professional domains.

Introduction

Recent advancements in large language models have achieved significant breakthroughs ranging from text generation to complex reasoning (Plaat et al. 2024; DeepSeek-AI et al. 2025). Although LLMs have achieved strong performance across general domains, their application to historical tasks remains a challenge. Historical text generation demands both factual accuracy and the ability to infer complex relationships from fragmented evidence (Cao et al. 2024).However, existing models often suffer from hallucinations (generating factually inaccurate content), particularly under the misleading prompts (Huang et al. 2025; Chen et al. 2024a), which limits their reliability in real-world historical analysis.

To mitigate these limitations, existing methods to enhance LLMs—such as reinforcement learning from human feedback (RLHF) (Ouyang et al. 2022) and chain-of-thought (CoT) fine-tuning(Chen et al. 2024b; Kim et al. 2023). These approaches primarily target general-purpose tasks, focusing on aligning model outputs with human preferences or emulating human-like reasoning processes. However, they exhibit notable limitations when applied to domains with high factual precision requirements. For instance, RLHF often lacks domain-specific factual constraints, resulting in outputs that

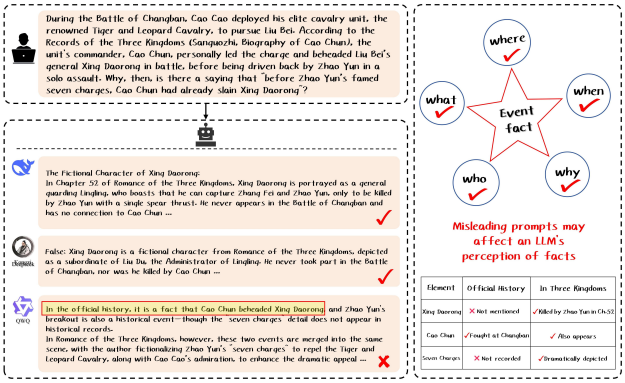

Figure 1: Comparative analysis of different LLM's responses to misleading historical fact-checking queries.

may be fluent and preference-aligned, but lack logical consistency with verified facts. Similarly, while CoT fine-tuning enhances reasoning ability, it does not inherently prevent the generation of factually inaccurate content. These limitations underscore the need for a domain-adaptive training framework for the historical domain—where factual reliability and deep, context-aware reasoning are essential.

Therefore, to address this issue, we propose a reasoning-oriented large language model specifically tailored for historical analysis tasks—**Kongzi**. By enhancing the model's understanding of historical facts, Kongzi is able to better analyze and answer historical questions, while achieving superior factual accuracy and consistency.

Specifically, we first collect domain-specific corpora related to history and conduct continued pre-training on top of a base model. We then perform instruction-following fine-tuning via a cold-start strategy using carefully constructed instruction data. Finally, we introduce a novel reward function: by building a graph of relevant information (entities and relations) and measuring its similarity to the ground truth, we use this as a reward signal to minimize the discrepancy between model output and factual references, thus improving the factual awareness of the model.We conduct comprehensive evaluations on public benchmarks as well as expert assessments. The results confirm the robustness of the proposed model and the effectiveness of our approach.

In summary, our contributions can be stated as follows:

- Continued Pre-training: We curated a diverse corpus of historical and conducted continued pre-trainingg to adapt the base model to the historical domain. This step ensures enhanced domain-specific knowledge for smaller models.
- CoT Data Generation: We classified initial data into factual statements and reasoning tasks, generating high-quality CoT data to enhance the model's reasoning ability through two-stage supervised fine-tuning (SFT). This approach enables the model to perform deep contextual reasoning, a critical capability for historical applications.
- Fact-Aware RL: We proposed a novel reward function that incorporates factual rewards into the reinforcement learning framework, ensuring that the model generates content aligned with historical facts.

Related Work

Reasoning in Large Language Models

In recent years, large reasoning models have shown incredible capabilities in a series of complex tasks, enabling them to solve complex tasks by decomposing problems into logical steps. Early reasoning approaches primarily relied on prompt engineering and few-shot learning (Yu et al. 2023), guiding models to generate intermediate reasoning steps through carefully designed prompts. While these methods excelled in handling simple tasks, they often struggled with more complex problems due to their dependence on static prompts and limited context. Recent research has explored dynamic reasoning frameworks, such as self-consistent decoding (Wang et al. 2023) and scratchpad prompting (Nye et al. 2021), which allow models to iteratively refine their reasoning processes.

Chain-of-Thought fine-tuning has emerged as a crucial technique for enhancing the reasoning capabilities of LLMs (Kim et al. 2023). By generating intermediate reasoning steps, CoT enables models to break down complex problems into simpler subtasks, resulting in more accurate and interpretable solutions. For example, zero-shot CoT demonstrated that LLMs can generate reasoning chains even without explicit fine-tuning, while CoT distillation (Wadhwa, Amir, and Wallace 2024) showed that high-quality CoT data can significantly boost model performance. (Yang et al. 2025).

Reinforcement Learning for Reasoning. Process-based reward models guide the model to perform structured problem-solving, while RL methods optimize performance through feedback-driven learning. Although these methods have achieved incremental gains, they still fall short of the general reasoning capabilities demonstrated by the o1. Recently, the DeepSeek-R1 model (DeepSeek-AI et al. 2025) has emerged as a competitive alternative, utilizing RL with the Group Relative Policy Optimization (Ramesh et al. 2024) algorithm, but the reward function it uses is only suitable for mathematical and coding tasks. Recent studies (Rafailov et al. 2024; Schulman et al. 2017; Zhong et al. 2025; Ramesh et al. 2024)have explored fact-aware RL where models are rewarded for generating factually accurate content, but these approaches typically focus on single-turn tasks and do not address the challenges of lengthy historical narratives. Our work introduces a novel fact-aware RL framework that evaluates the accuracy of each generated segment, ensuring consistent factual correctness throughout the generation process.

Chinese History in Large Language Models

Recent work has begun to explore large pretrained language models (LLMs) for multimodal understanding and generation within Chinese cultural contexts. For example, (Li et al. 2024) introduce the FoodieQA dataset, and (Zhao et al. 2025) propose the Fuxi, a systematic assessment framework that considers Chinese cultural authenticity.

In modeling and question–answering over Chinese historical knowledge, several studies have combined LLMs with structured knowledge bases to enhance factual accuracy and response quality. Despite advances in factual retrieval and knowledge integration, systematic evaluation of LLM temporal reasoning and multi-entity alignment remains scarce. The CTM benchmark (Wang et al. 2025) was proposed to address this gap, which constructs QA pairs and Timeline Ito Games that rigorously assess the abilities of models in cross-dynastic entity ordering, event alignment, and reasoning. In our work, we build on these insights by designing a history-enhanced reward function for reinforcement learning, and investigate how external knowledge injection can further optimize the temporal reasoning and alignment capabilities of models.

Method

To improve the factual accuracy of large models in the historical domain, we first collected and cleaned vertical domain data, then performed incremental pre-training on the pre-trained model to enhance its understanding of relevant information. Next, we conducted cold-start instruction fine-tuning to optimize the response structure, and finally applied a fact-enhanced reinforcement learning algorithm for further training. This approach effectively reduced the model's hallucination and significantly improved its performance.

Data acquisition and processing

Considering the continuity of ancient Chinese history and the completeness of its historical records, we selected it as the primary source of historical text data. With the assistance of domain experts, we effectively collected approximately 0.2 billion tokens of high-quality Chinese historical texts. Continued pre-training data: Data cleaning such as reducing noise and deduplication of the original corpus of 0.2B tokens SFT data:

(1) Classification: Historical corpora can be objectively categorised into two types: factual statements and contextual reasoning texts. The latter are used for subsequent data generation purposes. We employ a small-parameter LLM-assisted workflow for this separation. Specifically, when determining contextual information, the model segments text into multiple sentences based on punctuation marks, assigning each sentence a unique identifier. The incorporation of adjacent sentences as auxiliary context is necessitated by the potential for isolated sentences to lack semantic coherence and consequently lead to misjudgement. The remaining content is designated as factual information.

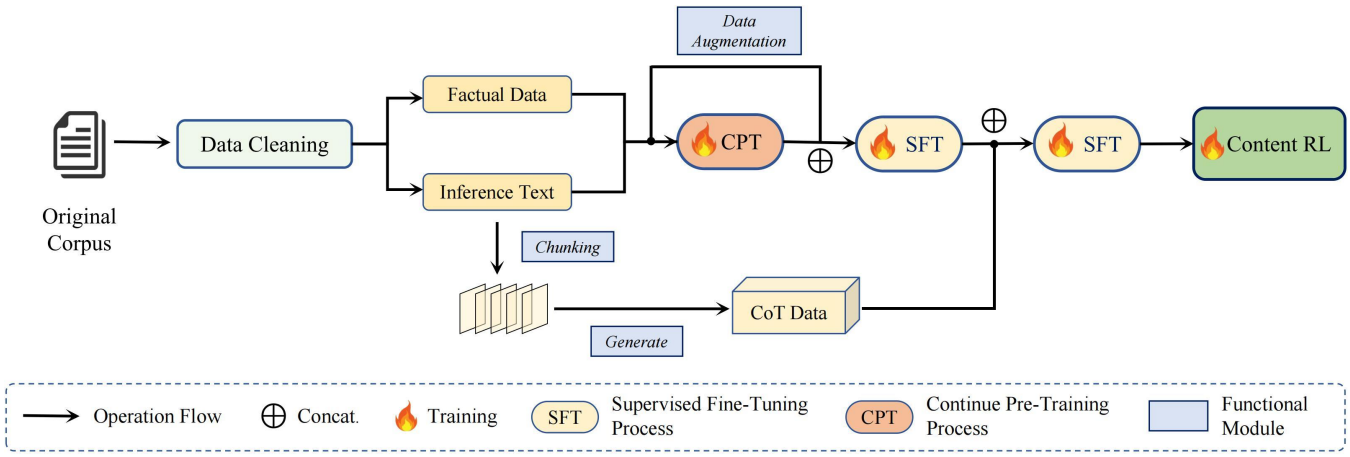

Figure 2: **Overview of our pipeline.** In terms of data processing, we clean the original corpus and separate it into factual data and inference text, where factual data undergoes augmentation while inference text is chunked for generating high-quality CoT data. For model training, we first conduct CPT with basic data, followed by a two-stage SFT to develop both fundamental QA capabilities and CoT reasoning abilities. Finally, Content RL is employed to optimize output quality through reinforcement learning.

(2) Generation of Primary SFT Data: To enhance the model's fundamental question-answering capability, we employ a random sampling method to construct simple question-and-answer pairs via LLM.

(3) Generation of Secondary SFT Data: The incorporation of chain-of-thought (CoT) data is imperative to facilitate the model's reasoning capacity. By leveraging the classified contextual reasoning texts, we guide the reasoning model to perform step-by-step analysis on each sentence and its contextual content. The output of this process follows a standardised CoT format, systematically constructing reasoning chain datasets.

Initial training strategy for the model

In the Continued pre-training (CPT) stage, the model has already obtained relevant historical information, but its command question-answering ability has not yet reached normal standards. To this end, we preliminarily optimize the pre-trained model through supervised command fine-tuning, so that it has preliminary command understanding and execution capabilities. To further enable the model to acquire reasoning capabilities, we introduce high-quality Chain-of-Thought data, and through the SFT method, enable the model to generate text according to the preset Chain-of-Thought template, thereby constructing a model architecture with preliminary reasoning capabilities, enabling it to have reasoning capabilities.

Reinforcement Learning Training

Previous studies aimed at enhancing the reasoning capabilities of language models have mainly adopted three strategies: first, using reward mechanisms to encourage models to output detailed reasoning processes; second, extending the model's reasoning time (i.e., increasing the number of output tokens); and third, using Retrieval-Augmented Generation (RAG) to address hallucinations caused by insufficient information through external knowledge bases. However, all three strategies have significant limitations: when the generated text becomes lengthy, the factual accuracy of the middle content becomes difficult to guarantee, making hallucinations more likely. Specifically, while language models can generate accurate short texts with clear prompts, they are more prone to factual errors or hallucinations in long-form generation, especially when context is limited or noisy data is introduced.

To address this, we propose a reinforcement learning-based optimization strategy. We introduce the Historical Fact-enhanced Reward Module (HFRM), which focuses on factual consistency by extracting entities and relations from generated text to build a candidate knowledge graph. This graph is compared with a ground truth graph, and their semantic similarity yields a factuality score that serves as part of the reward. This approach goes beyond surface-level evaluation to optimize the model's underlying knowledge structure.

Let

$$X = \frac{\sum_{i=1}^{G} \sum_{t=1}^{\|o_i\|} \mathcal{M}(r_{ratio}) \cdot \hat{A}_{i,t}}{G \sum_{i=1}^{G} \|o_i\|}, \tag{1}$$

$$\mathcal{M}(r) = \min\left[r, \mathrm{CLIP}(r, 1-\varepsilon, 1+\varepsilon)\right] \tag{2}$$

where:

- G is the size of the action group sampled for each state.
- o_i is the output sequence of the i-th action group.
- $\|o_i\|$ is the length of the output sequence of the i-th action group.
- r_{ratio} is the probability ratio of the current policy to the old policy.
- ε is the clipping parameter for the probability ratio.
- $\hat{A}_{i,t}$ is the relative advantage of the i-th action group at time step t.

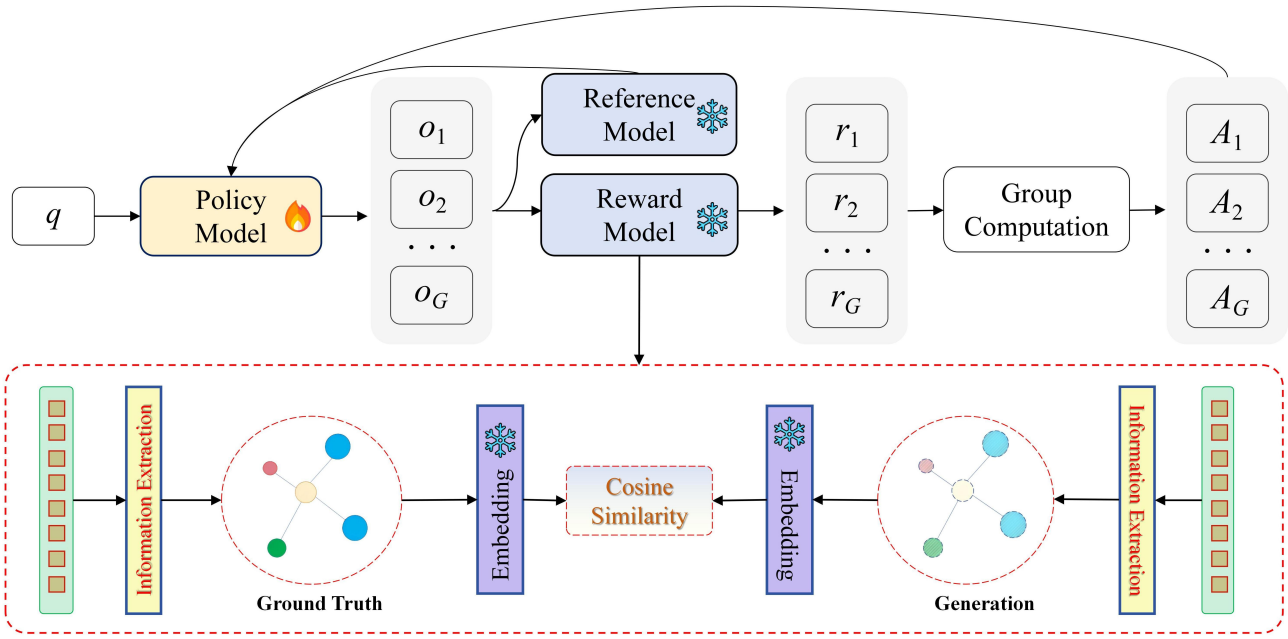

Figure 3: The HFRM module constructs a knowledge graph by extracting entities and relations from the model outputs, which serves as the foundation for information representation and computation. The semantic similarity between this graph and the ground truth is used as the reward signal.

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The objective function of Generalized Reward Policy Optimization (GRPO) can be expressed as:

$$\mathcal{J}^{\text{GRPO}}(\theta) = \mathbb{E}[X] - \varepsilon_{KL} \tag{3}$$

where:

- ε_{KL} is the KL divergence regularization term, used to constrain the difference between the new and old policies.

This objective function optimizes the policy model's parameters θ by minimizing the expectation of X, while introducing a KL divergence regularization term.

To enhance the factual consistency and semantic reliability of language models in knowledge-intensive tasks, we propose a Factuality-Oriented Reward Mechanism for Historical Entities. This mechanism aims to reinforce the factual accuracy and relational completeness of long-form text generation. It is built upon the traditional Generative Reinforcement with Policy Optimization (GRPO) framework, with the introduction of a novel component—Historical Fact-enhanced Reward Module (HFRM)—designed to more precisely capture the alignment between generated text and historical facts.

Specifically, we first perform information extraction on the generated text to identify entities such as historical figures, locations, and time points, as well as the relationships among them. Based on these, we construct a candidate knowledge graph corresponding to the generated content. This graph is then aligned with a ground truth knowledge graph curated by human experts. By calculating the semantic similarity between the two graphs—e.g., via knowledge graph embedding techniques—we derive a factuality score, which serves as one of the reward signals in the reinforcement learning process.

In terms of reward design, we adopt a fine-grained strategy: when the model correctly generates entities and relations that match the ground truth, a positive reward is given to encourage accurate learning and expression of historical facts. Conversely, if the generated output contains missing entities, incorrect recognition, distorted relations, or semantic inconsistencies (such as reversed relations or temporal logic errors), a negative reward is applied to suppress the generation of hallucinated or incoherent information.

By explicitly modeling factuality at the entity level, this mechanism not only enhances the model's sensitivity to historical content but also significantly mitigates hallucinations commonly observed in the generation of complex, multi-sentence long-form texts.

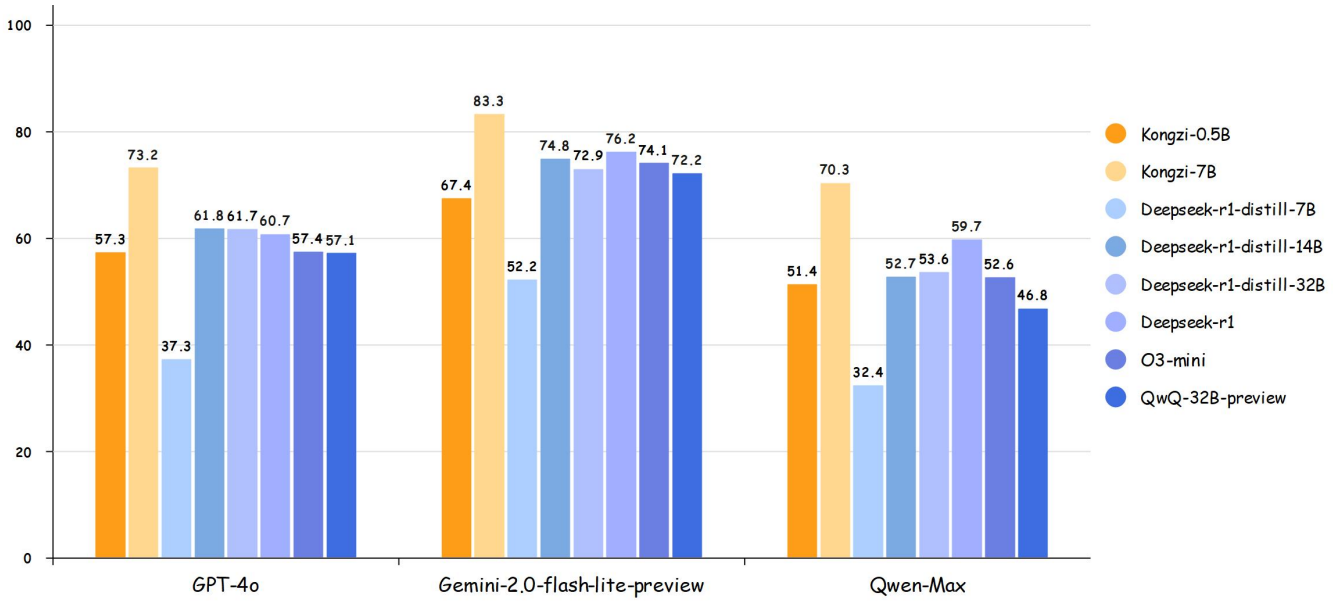

Figure 4: **Comparison of Kongzi's scores with other models under different judges.** The model score is a weighted sum of the following components: the score for the model's response process (the average score of Historical Accuracy, Logical Reasoning, and Problem Solving, with a weight of 0.8), the score for the thought process (with a weight of 0.1), and the ratio of the model's responses that outperform the Deepseek-r1 sample (with a weight of 0.1).

Experiments

Datasets and Evaluation Protocols

Given the relatively complete and continuous nature of ancient Chinese history, we have collected nearly 200 million tokens of Chinese historical and cultural annotated text corpus. This includes, but is not limited to, publicly available sources such as the Twenty-Four Histories. The corpus spans thousands of years of historical periods, events, and figures, ensuring a broad coverage throughout ancient Chinese history.

To facilitate targeted model training and obtain higher-quality data, we have objectively categorized the corpus into two types:

1. **Factual statement data:** Mainly contains objective descriptions of historical events, such as dates, places, and relationships between people.
2. **Contextual reasoning text:** Contains inferable value through contextual relationships, metaphors, and proverbs, such as analysis of historical causal relationships or discussion of the meaning of certain events.

Data Processing & Generating

Since most of the original texts are written in Classical Chinese, directly using them for continued pre-training may lead to semantic ambiguities in vernacular Chinese contexts. Therefore, after basic cleaning and preprocessing of the original data, we randomly sampled 700,000 entries and employed a large language model (LLM) to automatically generate question–answer pairs in both Classical and vernacular Chinese styles, thereby enhancing the training data for the subsequent supervised fine-tuning (SFT) stage.

For the factual historical data, we randomly selected and filtered 3,000 samples. The entities and relations in each sentence were extracted and then verified by domain experts.

For the contextual reasoning texts, we used an LLM to automatically identify 1,300 representative historical events and questions. Among them, 1,000 samples were constructed into structured instruction-tuning training data in Chain-of-Thought (CoT) format using Deepseek-R1, while the remaining 300 samples were used as a test set.

Base Model Training

We used Qwen2.5 (Qwen et al. 2025) with 0.5B and 7B parameters as the base model for our experiments.

Continued PreTraining We performed continued pre-training on the 0.5B and 7B models using cleaned classical Chinese data on the NVIDIA A6000 GPU for 1 and 5 days, respectively.

Supervised Fine-Tuning After continued pre-training, we conducted the first step of SFT on 1 million augmented data using eight NVIDIA GeForce RTX 3090 GPUs. To enhance the model's ability to answer historical questions while improving its general conversational capabilities, we used a combination of general datasets and our augmented dataset for SFT to achieve human alignment.

In the first step of the SFT process, we adopted a two-stage training strategy to balance general conversational ability and historical question-answering ability:

Model	Think Score	Answer Score	Historical Accuracy	Logical Reasoning	Problem Solving	>Deepseek-r1 Ratio
GPT-4o						
Kongzi-0.5B-SFT	50.42	45.68	33.87	56.59	46.59	52.18%
Kongzi-7B-SFT	60.46	64.27	50.97	76.96	64.88	85.29%
Kongzi-0.5B-RL	**60.42**$_{(+10.0)}$	**56.39**$_{(+10.7)}$	**53.34**$_{(+19.5)}$	**62.41**$_{(+5.9)}$	**53.43**$_{(+6.8)}$	**62.28**$_{(+10.1)\%}$
Kongzi-7B-RL	**72.34**$_{(+11.9)}$	**70.64**$_{(+6.4)}$	**58.45**$_{(+7.5)}$	**82.13**$_{(+5.2)}$	**71.34**$_{(+6.5)}$	**94.32**$_{(+9.0)\%}$
Gemini-2.5						
Kongzi-0.5B-SFT	70.89	61.78	54.61	64.38	66.36	42.17%
Kongzi-7B-SFT	80.13	74.72	68.39	76.04	79.72	57.83%
Kongzi-0.5B-RL	**77.91**$_{(+7.0)}$	**68.49**$_{(+6.7)}$	**61.13**$_{(+6.5)}$	**69.88**$_{(+5.5)}$	**74.45**$_{(+8.1)}$	**48.17**$_{(+6.0)\%}$
Kongzi-7B-RL	**89.24**$_{(+9.1)}$	**84.43**$_{(+9.7)}$	**80.45**$_{(+12.1)}$	**85.43**$_{(+9.4)}$	**89.42**$_{(+9.7)}$	**68.21**$_{(+10.4)\%}$
Qwen-Max						
Kongzi-0.5B-SFT	54.40	44.33	32.12	56.31	44.52	36.17%
Kongzi-7B-SFT	64.46	61.52	46.59	78.43	59.54	79.49%
Kongzi-0.5B-RL	**60.40**$_{(+6.0)}$	**50.49**$_{(+6.2)}$	**39.82**$_{(+7.7)}$	**67.12**$_{(+10.8)}$	**44.52**$_{(+0.0)}$	**49.18**$_{(+13.0)\%}$
Kongzi-7B-RL	**73.31**$_{(+8.9)}$	**67.82**$_{(+6.3)}$	**56.36**$_{(+9.8)}$	**84.78**$_{(+6.4)}$	**62.31**$_{(+2.8)}$	**87.21**$_{(+7.8)\%}$

Table 1: **Improvement of Kongzi model performance in various metrics by RL training.** We have listed the improvements of the RL-trained Kongzi 0.5B and 7B models over the SFT version across various metrics.

1. In the first stage, we performed SFT training using a general SFT dataset and half of our augmented dataset with block-level random mixing.
2. In the second stage, we used the other half of the augmented dataset for SFT.

caption

In the second-stage SFT, we fine-tuned the model on 1,000 Chain-of-Thought (CoT) examples for structured instruction following. This cold-start step enabled the model to learn self-reflection and autonomous reasoning from these long-form CoT trajectories, laying the groundwork for more rapid and effective acquisition of extended CoT capabilities in the subsequent RL phase.

Reinforcement Learning Training

After completing continued pre-training and SFT, we further optimized the model's performance through RL using 3000 data points.

Reinforcement Learning Framework We adopt GRPO as the reinforcement learning framework, incorporating a newly designed reward function to ensure factual accuracy during the model's generation process. In this reward function, we compute the cosine similarity between the knowledge graphs of the ground truth and the model output as the reward signal.

Specifically, we designed the following reward functions to assess the model's output:

- Formula:

$$ER = \cos\left(f(\mathcal{G}_{\text{model}}),\ f(\mathcal{G}_{\text{gold}})\right)$$

where $\mathcal{G}_{\text{model}}$ and $\mathcal{G}_{\text{gold}}$ denote the knowledge graphs extracted from the model output and the ground-truth, respectively, and $f(\cdot)$ is a graph embedding function that maps a knowledge graph to a dense vector representation.

1. **Format Reward:**
   - Check for format markers like `<|begin_of_thought|>`.
   - Reward: 1.0 for correct format, 0.0 otherwise.
2. **Repetition Penalty Reward:**
   - Penalize repeated n-grams with a negative value based on repetition ratio.
   - Formula:

$$RP = RR \times MPV$$

Integration of Reward Functions We combined multiple reward functions with weighted aggregation. The final reward is calculated as follows:

$$R_{\text{final}} = w_1 R_{\text{cos}} + w_2 R_{\text{log}} + w_3 R_{\text{fmt}} + w_4 R_{\text{rep}} \quad (4)$$

where:

- R_{cos}: Factual reward
- R_{log}: Logical coherence reward
- R_{fmt}: Format reward
- R_{rep}: Repetition penalty reward
- w_1, w_2, w_3, w_4: Corresponding weights, where $\sum_{i=1}^{4} w_i = 1$

Base Model Table			
Category	**Model**	**w/o**	**Cot**
Dense Model	Qwen2.5-72B	**91.02%**	—
Dense Model	Qwen2.5-7B few-shot	87.31%	69.88%
Dense Model	Qwen2.5-0.5B few-shot	35.29%	29.41%
Mixed Reasoning Model Table			
Category	**Model**	**w/o**	**Thinking**
Mixed	Qwen3-235B-a33B few-shot	**92.88%**	—
Mixed	Qwen3-30B-a3B few-shot	89.47%	—
Mixed	Qwen3-32B few-shot	90.40%	—
Reasoning Model Table			
Category	**Model**	**Reasoning**	—
Reasoning	Deepseek r1	**92.88%**	—
Reasoning	Deepseek r1 distill 7B	40.56%	—
Reasoning	Deepseek r1 distill 14B	81.11%	—
Reasoning	Deepseek r1 distill 32B	86.69%	—
Reasoning	QwQ-32B	—	—
Reasoning	Kongzi-0.5B	59.37%	—
Reasoning	Kongzi-7B	87.56%	—
Reasoning	Kongzi-0.5B (RL)	71.21%	—
Reasoning	Kongzi-7B (RL)	90.37%	—

Table 2: Results on CMMLU-Chinese-History. Best scores in each metric are highlighted in bold. 'w/o' refers to accuracy without enhancement prompts (e.g., Cot or Thinking). '—' indicates the model was not tested or data is unavailable under the corresponding condition.

Results Evaluation

To ensure fairness and reasonableness,we conducted evaluations on two test sets: the publicly available Chinese historical benchmark CMMLU, and our self-constructed test set containing 300 samples. we used both manual expert scoring(Twelve volunteers voted on the responses of each model, and in the end, Kongzi-7B's response received 72.13% support.) and LLM scoring. Due to the different knowledge and preferences of various LLMs (Gu et al. 2025), we selected three mainstream LLMs(GPT-4o (OpenAI et al. 2024), Gemini-2.0-flash-lite-preview,Qwen-Max) to score according to our scoring template:

(1) Think Score: The score of thinking ability or strategy.

(2) Answer Score: The score of the quality of the answer result.

(3) Historical Accuracy: The accuracy of the statement of real historical events.

(4) Logical Reasoning: Whether the logic of reasoning is correct.

(5) Problem Solving: Whether the problem is solved normally.

The results are shown in Figure 4 and Table 1. The results demonstrate that our Kongzi model has excellent capabilities in all aspects, with the 7B model surpassing the Deepseek-R1 model and achieving the highest score. It is also worth mentioning that the Kongzi 0.5B model has shown astonishing inference capabilities, outperforming DeepSeek-r1-Distill-7B and 14B.

To demonstrate the effectiveness of our RL training, we present a comparison of the scores of the RL version of the Kongzi 0.5B model and the 7B model with the SFT version. The results are shown in Table 2 and 300 samples results are shown in Table 1. The results demonstrate that the RL-trained models achieved significant improvements in all metrics.

Conclusion

In this paper, we propose Kongzi, a language model specifically designed for the historical domain. Our work begins with continued pre-training on historical corpora to enhance domain-specific knowledge within a small-parameter model. Then, we conduct an effective cold-start training process to aligns the model with the question-answering format, enabling it to better adapt to downstream historical tasks. Finally, we adopt the graph-enhanced reinforcement learning method to reduce hallucinations arising in long-chain reasoning and misleading prompt scenarios, while further improving the model's reasoning ability. Extensive experiments on multiple historical benchmarks demonstrate that Kongzi, despite its relatively small parameter size, consistently achieves strong performance across a wide range of tasks.

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